



Name: Faizan Akram

Reg No: BCSP 243006

Assignment No. 1

Submitted to: Ma'am Muntaha Tabassum

Assignment 1

Question 1:

In an area where no internet, no router & no access point is available, a group of students creates two types of network:

- **Wifi** (Ad hoc mode)

- **Bluetooth** (Piconet)

After testing, they claim:

"Both networks behave the same because both are wireless & do not use infrastructure."

(1)

Is this statement correct or incorrect? Justify the answer with logical reasoning?

The statement is incorrect. Although both Wifi and Bluetooth are wireless & infrastructureless, but they do not behave the same.

Their behaviour & working are significantly different.

Logical Reasons:

1) Network Structure & Control

⇒ In Wifi Ad hoc mode, all devices act as equal peers & communicate directly without any central controller.

⇒ In Bluetooth Piconet, there is a master-slave structure where one device controls communications and others follow it. This creates different communication behavior.

2) Range & Coverage :

- ⇒ Wifi Ad hoc networks typically support a larger range (10^{meters} to more depending on Power).
- ⇒ Bluetooth Piconets have a very short range (usually around 10 meters), making them suitable only for close proximity communication.

3) Data Transfer Speed :

- ⇒ Wifi supports much higher data rates, allowing activities like file sharing or streaming.
- ⇒ Bluetooth is designed for low Power and lower speed communication, so performance is not the same.

4) Number of Devices Supported :

- ⇒ Wifi Ad hoc can support multiple devices flexibly.
- ⇒ Bluetooth Piconet is limited (typically 1 master and upto 7 active slave devices), restricting scalability.

5) Power Consumption Behavior :

- ⇒ Wifi consumes more Power and is less efficient for small devices.
- ⇒ Bluetooth is optimized for low Power usage, which changes how devices behave in long term communication.

(2)

Can Both Networks support TCP/IP communication in the same way? Explain

No, Both Network cannot support TCP/IP communication in the same way.

- Wifi Ad hoc mode is designed to work manually with TCP/IP Protocols, just like regular networks. Devices can easily assign IP address and communicate using standard networking applications.
- Bluetooth requires additional Protocol layers (like Bluetooth stack and Profiles) to support TCP/IP. It is not inherently designed for full TCP/IP networking and may have limitations or require special configuration.

Wifi

- Fully support TCP/IP Protocols directly
- Devices can communicate like normal network (Just without a router)
- Used for file sharing, gaming etc...

Bluetooth

- Does not Natively use TCP/IP.
- Use its own Protocol Bluetooth stack
- TCP/IP can be used only with extra configuration and even then it is limited.

← The End →